Clinical Treatment

Sepsis ICU Management with Reinforcement Learning

Reinforcement Learning Course Project

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1. Introduction

1.1 The Clinical Problem

Sepsis is one of the leading causes of mortality in intensive care units (ICUs) worldwide, responsible for millions of deaths annually. It is a life-threatening systemic response to infection that requires continuous, high-stakes clinical decisions: how much vasopressor medication to administer to stabilise blood pressure, and how much intravenous (IV) fluid to give to maintain circulation. These two interventions must be carefully balanced — too little risks patient deterioration, while too many risks iatrogenic harm.

Traditionally, these decisions are made by ICU clinicians relying on clinical guidelines and experience. However, the complexity of the problem due to the hundreds of possible patient states, dozens of treatment combinations, delayed and sparse outcome signals, and high inter-patient variability, makes it a natural candidate for Reinforcement Learning. A well-trained RL agent can learn treatment policies directly from patient data, potentially uncovering strategies that improve survival rates while minimising unnecessary intervention.

1.2 Why Reinforcement Learning?

Sepsis management maps directly onto the RL framework: the patient's physiological state is the observation, vasopressor and IV fluid dosing combinations are the actions, and survival is the terminal reward (+1 in case of survival and 0 for fatalities). The sequential, delayed nature of the reward signal, where treatment decisions made hours earlier affect eventual outcomes, makes supervised learning inadequate and RL the natural approach. The agent must learn to balance exploration of treatment strategies against exploitation of known effective policies, mirroring the real clinical tension between established protocols and individualised care.

1.3 Project Overview

This project develops and evaluates RL agents on the ICU-Sepsis-v2 benchmark, constructed from the real MIMIC-III clinical database. The project is structured around two configurations of increasing complexity, and an extra configuration altering the reward system:

Configuration A: A discrete state space of 716 states and 25 actions, enabling tabular RL and model-based dynamic programming. We implemented Policy Iteration (PI) and Q-Learning, using PI as the theoretical performance ceiling, given that it computes exact value functions via Bellman updates and improves the policy greedily. As such, it is guaranteed to converge to the optimal policy in finite iterations for finite MDPs.

Configuration B: A 47-dimensional continuous physiological feature vector, rendering tabular methods infeasible and requiring deep RL. Three clinical reality wrappers inject observation noise, missing lab values, and acute patient events. We implemented DQN (Deep Q-Learning), PPO (Proximal Policy Optimisation), and A2C (Advantage Actor-Critic).

Creative Extension: A SOFA-based reward shaping mechanism with systematic sensitivity analysis, revealing an important interaction between reward shaping and robustness under partial observability.

2. Configuration A

2.1 Environment Description

The ICU-Sepsis-v2 environment in Configuration A presents the patient state as a discrete integer index derived from discretising 47 clinical variables into 716 states. The 25 actions correspond to a 5×5 grid of vasopressor levels (0–4) and IV fluid doses (0–4). The reward signal is sparse: +1 at survival, 0 at death, and also 0 for all intermediate transitions, with discount factor γ = 1.0, meaning it considers considers future rewards as much as immediate ones. A treatment intensity penalty (λ = 0.02) and SOFA score bias of 5.0 are active, producing a mean starting SOFA of 9.02. The full MDP model, transition matrix P(s'|s,a) of shape (716 × 25 × 716) and reward matrix R(s,a) are directly accessible, enabling model-based algorithms.

2.2 Algorithms

Policy Iteration

Policy Iteration (PI) alternates between policy evaluation — computing the exact value function V(s) via iterative Bellman updates with convergence threshold θ = 1e-10 — and policy improvement — greedily updating the policy from derived Q-values. Terminal state values are pinned throughout: V(survived) = 1, V(died) = 0. PI requires no environment interaction and is guaranteed to converge to the optimal policy for finite MDPs in finite iterations. It serves as the performance ceiling for all model-free Config A methods. PI is the natural choice when transition dynamics are fully known: its convergence to the exact optimal policy makes it the ground truth for Config A evaluation.

Q-Learning

Q-Learning, in opposition, is an off-policy temporal difference algorithm. The update rule consists on: $Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \cdot \max_{a'} Q(s',a') - Q(s,a)]$. The bootstrapped greedy target makes it off-policy: it learns the optimal policy regardless of the behaviour policy used for exploration. We implemented three variants of increasing sophistication, aiming for a higher survival rate:

Hyperparameter	Baseline	Improved	Final
Initialisation	0	1	1
Learning Rate	Fixed 0.1	Visit-based 1/n(s,a)	Visit-based 1/n(s,a)
ε start	1	1	1
ε end	0.01	0.05	0.05
Episodes	75,000	200,000	400,000
Tie-Breaking	No	Yes	Yes

Table 1: Q-Learning hyperparameters across variants.

Optimistic initialisation ($Q_0 = 1.0$) treats unvisited state-action pairs as maximally promising, driving the agent to explore them before settling — an intrinsic exploration bonus. Visit-based learning rates (α = 1/n(s,a)) satisfy the Robbins-Monro convergence conditions, providing theoretical convergence guarantees absent with fixed α. Extended training and a higher ε_end allow more complete state-action coverage before exploitation dominates.

Comparing Policy Iteration and Q-Learning provides insight into the trade-offs between model-based and model-free reinforcement learning. Although both operate in the same discrete environment, PI exploits complete knowledge of the transition dynamics to compute the optimal policy directly, while Q-Learning must discover effective behaviour solely through interaction and exploration. Evaluating both allows us to quantify the cost of learning without a model, compare sample efficiency, and establish an optimal performance benchmark.

2.3 Results

2.3.1 Policy Iteration and Baselines

The random agent achieves 69.3% survival, establishing the performance floor. Policy Iteration converges in 3 iterations (Figure 1), achieving 84.1% survival, the theoretical optimum for this MDP. The 14.8 percentage point gap between random baseline and PI ceiling defines the improvement range for model-free methods. PI's rapid convergence (seconds of compute, no environment interaction) illustrates the power of model-based methods when dynamics are known exactly.

Method	Survival Rate	Mean Return	Notes
Random baseline	69.3%	0.693	Performance floor
Policy Iteration	84.1%	0.841	Theoretical ceiling (model-based)
QL Baseline	76.1%	0.761	Zero init, α=0.1, 75k episodes
QL Improved	79.4%	0.794	Optimistic init, visit-α, 200k episodes
QL Final (tuned)	82.5%	0.825	ε_end=0.05, 400k episodes — best model-free

Table 2: Configuration A results summary (1,000 evaluation episodes).

2.3.2 Q-Learning Analysis

The baseline Q-Learning agent (76.1%) substantially improves on the random policy (+6.8 pp), confirming that tabular RL is effective in this environment. The improved variant (79.4%) gains a further 3.3 pp through optimistic initialisation, visit-based learning rates, and tie-breaking. The final tuned variant (82.5%) approaches within 1.6 pp of the PI ceiling after hyperparameter optimisation.

2.3.3 PI vs Q-Learning Comparison

The results demonstrate a clear progression in performance across the Configuration A methods. As illustrated in Figure 2, survival increases from 69.3% for the random baseline to 76.1% for baseline Q-Learning, 79.4% for the improved variant, 82.5% for the final tuned variant, and 84.1% for Policy Iteration. This stepwise improvement indicates that each algorithmic enhancement the optimistic initialisation, visit-based learning rates, and extended training contribute incrementally toward the optimal Policy Iteration solution. The remaining 1.6 percentage point gap between the final Q-Learning policy and Policy Iteration

represents the residual cost of model-free learning after all improvements have been applied.

The relationship between Policy Iteration agreement and survival, shown in Figure 3, reinforces this conclusion. Q-Learning variants that agree with the optimal Policy Iteration policy in a larger proportion of non-terminal states consistently achieve higher survival rates. The positive correlation indicates that performance improvements are driven by recovering the optimal treatment decisions rather than by chance. Disagreement with the Policy Iteration policy therefore provides a direct explanation for the remaining performance gap.

The learning curves in Figure 4 reveal important differences in training behaviour. Baseline Q-Learning exhibits substantial variance during the early stages of training because unvisited state-action pairs are initialised to zero, leading to arbitrary action selection and noisy return estimates. In contrast, the improved and final variants produce smoother learning trajectories. Optimistic initialisation encourages systematic exploration of previously unvisited state-action pairs, allowing the Q-table to be populated more effectively before exploitation dominates. The final variant continues to improve beyond the point where the improved variant plateaus, demonstrating the benefit of extended training and more complete state-space coverage.

Additional insight is provided by the treatment-intensity analysis in Figure 5. Policy Iteration achieves the highest survival rate while administering less treatment on average than the random policy, indicating that successful treatment depends on selecting appropriate interventions rather than simply increasing intervention intensity. The Q-Learning variants lie between the random and optimal policies on both survival and treatment intensity, progressively converging toward the Policy Iteration treatment profile as performance improves. This suggests that the agents are discovering more effective treatment strategies rather than surviving patients through excessive intervention.

The exploration–exploitation schedules visualised in Figure 6 help explain these performance differences. The baseline agent reduces exploration relatively quickly and consequently converges to a lower-performing policy. Extending exploration in the improved and final variants allows the agent to continue discovering beneficial state-action pairs before exploitation becomes dominant, leading to progressively higher survival rates. The results demonstrate that sustained exploration is essential for identifying effective treatment strategies and avoiding premature convergence to suboptimal policies.

Finally, the action distributions in Figure 7 provide a direct view of the learned treatment strategies. Policy Iteration concentrates decisions on a small subset of treatment actions, particularly actions 0, 5, 10, 15, and 20, indicating that only a limited number of intervention combinations are optimal across the state space. The final Q-Learning policy recovers the same overall structure, assigning many states to these dominant actions. However, it distributes decisions across a broader range of treatments than Policy Iteration, reflecting residual inaccuracies in its value estimates. These differences account for the remaining 1.6 percentage point survival gap and demonstrate that the final Q-Learning policy has learned much of the structure of the optimal treatment strategy without fully recovering it.

3. Configuration B

3.1 Environment Description

Configuration B provides a 47-dimensional continuous feature vector of normalised physiological measurements, including SOFA score, heart rate, lactate, blood pressure, creatinine. With continuous observations, a tabular Q-table is infeasible as the probability of revisiting an identical observation vector is effectively zero, so a neural network function approximation becomes necessary.

Three clinical reality wrappers add orthogonal failure modes reflecting real challenges:

- EpisodicNoisyObsEnv: With probability 0.15, all observations in an episode are corrupted by Gaussian noise ($\sigma = 0.10$), modelling monitor malfunction or calibration drift.
- EpisodicMissingObsEnv: With probability 0.15, a random subset of 4 features is zeroed out for the entire episode, modelling lab analyser downtime or missing chart entries at admission.
- AcuteEventEnv: At each step, with probability 0.01, the patient suffers an irreversible acute event (cardiac arrest, pulmonary embolism) and dies regardless of treatment. Over a typical 9-step episode this produces approximately 9% of episodes with at least one acute.

3.2 Algorithms

With a 47-dimensional continuous observation, tabular methods are infeasible. All three algorithms use a [128, 128] MLP architecture. Key differences are summarised below:

Hyperparameter	DQN	PPO	A2C
Type	Off-policy	On-policy	On-policy
Exploration	ε-greedy (decay 0.4×1M)	Entropy coef 0.01	Entropy coef 0.01
Learning rate	1e-3	3e-4	7e-4
Rollout / buffer	Buffer 100k	n_steps=512, 4 envs	n_steps=128, 4 envs
Batch size	64	64	—
Update epochs	1 per step	10 per rollout	1 per rollout
Target network	Every 500 steps	—	—
Timesteps	1,000,000	2,000,000	2,000,000
Final survival rate	65.0%	66.2%	71.0%

Table 3: Configuration B algorithm hyperparameters and final survival rates.

DQN was selected as the value-based baseline. Experience replay reduces correlation between consecutive observations and improves sample efficiency; the target network stabilises learning by providing slowly changing bootstrap targets. Its exploration

mechanism is strictly ε-greedy: exploration is uniformly random during the decay phase and then fixed at ε = 0.05. This is computationally cheap but inflexible — once ε is low, the agent exploits its current (potentially incorrect) estimates with no mechanism to revisit them.

PPO was selected as a modern on-policy policy-gradient algorithm. It directly optimises a stochastic policy and maintains exploration throughout training via an entropy coefficient (ent_coef = 0.01), which adds a bonus proportional to the policy's action distribution entropy. The clipped surrogate objective constrains update size, preventing destabilising policy changes. Longer rollout trajectories (n_steps = 512 per environment, 4 environments = 2,048 transitions per update) allow PPO to collect diverse patient experiences before updating, which is expected to improve generalisation under the stochastic clinical wrappers.

A2C was selected to assess whether shorter rollouts and more frequent updates could outperform PPO in the sparse-reward sepsis setting. A2C performs synchronous updates from n_steps = 128 per environment across 4 environments (512 transitions per update). Like PPO, it maintains entropy regularisation. Unlike PPO, it lacks clipping and multi-epoch optimisation, making each update simpler but potentially noisier. The three algorithms represent the full spectrum from pure value-based (DQN) to simple on-policy (A2C) to stable on-policy (PPO), allowing a systematic investigation of exploration mechanisms and update frequency under clinical constraints.

3.3 Configuration B Results

3.3.1 Training Curves and Convergence Speed

Figure 8 shows the smoothed training return curves (MA200) for all three algorithms over their respective training budgets. The substantial overlap between the curves indicates that learning rates are broadly similar, making it difficult to draw strong conclusions about convergence speed from the training curves alone. None of the algorithms exhibit instability or catastrophic performance collapse, suggesting that the selected hyperparameters provide stable optimisation across all methods.

A key limitation when interpreting these curves is that DQN was trained for 1 million timesteps, whereas PPO and A2C were trained for 2 million timesteps. This difference was primarily due to computational constraints, as DQN required considerably longer wall-clock training time than the actor-critic methods. Consequently, the final performance comparison is not based on an identical interaction budget. Some of the observed performance gap between DQN and the actor-critic algorithms may therefore be attributable to the reduced training budget rather than algorithmic differences alone. Additional training could potentially allow DQN to further refine its value estimates and close part of the performance gap.

Because the training curves overlap substantially, final evaluation metrics provide a more reliable basis for comparison. Under greedy evaluation, A2C achieved the highest overall performance, followed closely by PPO, while DQN obtained the lowest mean return. Action entropy analysis further revealed important behavioural differences that are not visible in the training curves. PPO maintained the most diverse treatment policy, consistent with its entropy-regularised exploration mechanism, whereas DQN converged to a more concentrated action distribution. These findings suggest that the primary distinctions between the algorithms emerge not from dramatically different learning rates but from

differences in the quality, diversity, and robustness of the policies learned by the end of training.

Figure 9 shows smoothed survival rate curves (MA200), where, once again, the behaviour was rather similar among the three algorithms.

3.3.2 Action Distribution and Exploration vs Exploitation

Figure 10 shows the action distribution heatmaps (5×5 grids where rows are vasopressor levels 0–4 and columns are IV fluid levels 0–4) for all three algorithms at evaluation time. These heatmaps directly visualise the exploitation outcome of each exploration strategy: the actions a deterministic greedy policy selects across 1,000 evaluation episodes.

DQN's heatmap shows a near-complete concentration on a single cell: vasopressor level 1, IV fluid level 3, selected in approximately 97% of all steps. Action entropy is 0.178 nats, just 5.5% of the theoretical maximum ln(25) = 3.219. This is the direct consequence of ε-greedy exploration's structural limitation in sparse reward environments. Once ε is low, the agent exploits one action with near certainty. In the clinical context, this is equivalent to a clinician giving every ICU patient the same vasopressor and fluid combination regardless of their individual physiological state, a treatment strategy that ignores the 47-dimensional observation entirely. The fact that DQN's survival rate (65.0%) barely exceeds the random baseline (64.6%) confirms that this uniform strategy provides negligible clinical benefit.

PPO's heatmap shows high diversity across all 25 action cells, with no single action dominating. Action entropy is 2.872 nats, 89.2% of maximum, close to a uniform distribution over treatments. This is the direct consequence of entropy regularisation: the ent_coef = 0.01 penalty prevents the policy from collapsing to a single action even after convergence. Clinically, PPO maintains the most treatment diversity, assigning different vasopressor-fluid combinations to different patient states. However, high entropy alone does not guarantee good performance since a near-uniform policy is nearly identical to random action selection in its clinical effect. PPO's 66.2% survival suggests its diverse policy has learned some state-dependent structure, but the entropy bonus may also be preventing it from fully exploiting the treatment combinations it has identified as most effective.

A2C's heatmap occupies a clinically meaningful middle ground. Action entropy is 1.461 nats (45.4% of maximum), with probability concentrated in the low-vasopressor, low-IV-fluid quadrant of the 5×5 grid. This reflects a policy that has converged to a specific treatment philosophy with a conservative dosing as the default, while retaining enough diversity to adapt to different patient states. Critically, this concentrated-but-not-collapsed distribution aligns with clinical best practice in sepsis management: conservative fluid resuscitation and low-dose vasopressor support are generally preferred over aggressive intervention, with escalation reserved for deteriorating patients. A2C's action distribution suggests the agent has discovered this clinically appropriate structure from the reward signal alone, without any explicit encoding of clinical guidelines.

Algorithm	Action Entropy (nats)	% of Maximum	Dominant treatment	Clinical interpretation
DQN 1M	0.178	5.5%	VP=1, IV=3 (97% of steps)	Near-uniform: ignores patient state

Algorithm	Action Entropy (nats)	% of Maximum	Dominant treatment	Clinical interpretation
A2C 2M	1.461	45.4%	Low VP, low IV quadrant	Conservative, state-adapted policy
PPO 2M	2.872	89.2%	No dominant action	Diverse: state-dependent but diffuse

Table 4: Action entropy at evaluation. Maximum entropy = ln(25) = 3.219. VP = vasopressor level, IV = IV fluid level.

The exploration-exploitation trade-off in Config B can be framed as follows (Figure 11). DQN's ε-greedy strategy successfully explores during the decay phase but commits entirely to exploitation once ε is fixed, where exploration is switched off. This works well in dense-reward environments where the agent has seen enough transitions to accurately estimate Q-values but fails in sparse-reward environments like sepsis where many state-action pairs have never been visited when exploitation begins. The result is a policy locked onto whatever action happened to produce the first survival reward, which need not be optimal.

PPO and A2C solve this by never fully switching off exploration: the entropy coefficient maintains a permanent exploration incentive proportional to the policy's current concentration. As the policy becomes more concentrated (exploitation), the penalty increases, providing a restoring force toward diversity (exploration). This is an automatic, adaptive balance rather than a manually scheduled decay. The trade-off between PPO and A2C is then between stability (PPO's clipping and longer rollouts) and speed (A2C's more frequent updates). In the sepsis setting with sparse survival rewards, speed wins: A2C's faster incorporation of each survival signal produces a better-calibrated policy by the end of training.

The clinical implication of this exploration-exploitation analysis is significant. An agent that fully commits to exploitation too early, like DQN, risks adopting a rigid treatment protocol that fails to generalise across the heterogeneous ICU patient population. An agent that maintains appropriate exploration throughout training, like A2C, continues to discover which treatment combinations work for which patient states, producing a more personalised and ultimately more effective treatment policy.

3.3.3 Overall Performance Comparison

Table 5 and Figure 11 summarise the final performance of all algorithms evaluated over 1,000 episodes on the wrapped clinical environment.

Algorithm	Survival	Mean Return	Noisy Eps Return	Missing Eps Return	Acute Eps
Random baseline	64.6%	0.5610	—	—	8%
DQN 1M	65.0%	0.5558	0.5256	0.5455	8.9%
PPO 2M	66.2%	0.6068	0.5927	0.4933	8.4%
A2C 2M	71.0%	0.6818	0.6764	0.7170	7.0%

Table 5: Configuration B results summary (1,000 evaluation episodes). Noisy/Missing Eps columns show mean return on those episode subsets.

A2C achieves the best overall performance at 71.0% survival and mean return 0.6818, outperforming the random baseline by 6.4 percentage points. PPO achieves 66.2% survival, a modest improvement over random (+1.6 pp) but with substantially higher mean return (0.6068 vs 0.5610), reflecting that PPO's treatment decisions reduce unnecessary intervention even when survival outcomes are similar. DQN achieves 65.0% survival, barely exceeding the random baseline (+0.4 pp) — consistent with its near-deterministic collapsed policy.

The gap between A2C and PPO (4.8 pp survival) is the most practically significant finding in Config B. Both are on-policy actor-critic methods with identical network architectures, trained for the same number of timesteps. The difference is attributable primarily to update frequency: A2C's 128-step rollouts versus PPO's 512-step rollouts produce approximately 4× more gradient updates over the same training budget, allowing A2C to extract more learning signal from each rare survival event.

It is important to contextualise these results. The effective performance ceiling in Config B is substantially lower than Config A for two reasons: acute events (~9% of episodes) represent irreducible stochasticity where no policy can improve outcomes, and the continuous observation space requires generalisation across unseen states rather than memorisation of exact values. A2C's 71.0% survival, representing a 6.4 pp improvement over the random baseline, should be interpreted against this lower ceiling.

3.3.4 Robustness to Clinical Wrappers

Table 5 also breaks down algorithm performance across the three clinical failure modes.

Observation noise (EpisodicNoisyObsEnv): All algorithms show modest degradation under noisy episodes. DQN degrades from 0.5558 to 0.5256 (−5.4% relative), PPO from 0.6068 to 0.5927 (−2.3%), and A2C from 0.6818 to 0.6764 (−0.8%). A2C's near-negligible degradation reflects that its concentrated low-dose policy is inherently less sensitive to precise observation values — a conservative dosing strategy defaults to the same treatment region regardless of small measurement perturbations.

Missing observations (EpisodicMissingObsEnv): This wrapper reveals the starkest differences. PPO degrades dramatically from 0.6068 to 0.4933 (−18.8% relative), the largest degradation of any algorithm under any wrapper condition. PPO's diverse, state-dependent policy relies heavily on precise feature values to discriminate between patient states. When 4 of 47 features are zeroed, this discrimination fails. In contrast, A2C improves slightly under missing observations (0.6818 to 0.7170, +5.2%), a counterintuitive finding explained by A2C's concentrated policy: when the policy defaults to conservative low-dose treatment regardless of exact state, the removal of some features does not disrupt it. DQN shows moderate degradation (0.5558 to 0.5455, −1.9%), consistent with its near-observation-independent collapsed policy.

Acute events: The fraction of episodes with acute events is consistent across algorithms (7.0–8.9%), confirming these events are environment-driven rather than policy-driven. No algorithm can prevent acute events by design. The slight variation reflects random seed effects in the 1,000-episode evaluation.

The robustness analysis reveals a fundamental tension: PPO's diverse, state-dependent policy achieves better treatment personalisation under normal conditions but degrades

significantly when observations are incomplete. A2C's more conservative policy sacrifices treatment diversity in exchange for robustness under data unavailability. In a real clinical deployment where incomplete lab panels are routine, A2C's robustness profile is arguably more deployable.

3.3.4 Clean vs Wrapped Environment

To isolate the performance cost attributable to the three clinical wrappers, all three algorithms are retrained on the base continuous environment without EpisodicNoisyObsEnv, EpisodicMissingObsEnv, or AcuteEventEnv, using identical hyperparameters and training budgets. The clean random baseline is computed separately on the wrapper-free environment.

The wrapper cost table below shows survival rates on the clean environment alongside the wrapped results from Table 5, and the absolute performance cost of deploying in the full clinical setting:

Algorithm	Clean Survival	Wrapped Survival	Wrapper Cost
PPO 2M	74.0%	66.2%	8.2 pp
A2C 2M	71.6%	71.0%	0.6 pp
DQN 1M	70.4%	65.0%	5.4 pp

Table 6: Wrapper cost analysis

All three algorithms perform better on the clean environment, as expected: without observation noise, missing features, or acute events, the agents face a simpler, more stationary problem. The pattern of wrapper costs reveals important information about each algorithm's robustness profile.

PPO suffers the largest absolute degradation when moving from clean to wrapped. This is consistent with the robustness analysis in Section 3.3.4: PPO's diverse, state-dependent policy relies on precise feature values to discriminate between patient states, making it most sensitive to both observation noise and missing features.

A2C shows the smallest degradation of the on-policy methods, confirming that its concentrated low-dose policy is inherently robust to observation perturbations. Even on the clean environment, A2C achieves strong performance, and the gap to its wrapped performance is modest. This reinforces the finding from Section 3.3.4: A2C's conservative treatment strategy is not just a consequence of training on the wrapped environment but an emergent property that happens to be both effective and robust.

DQN' performance on a clean environment surpasses 70% survival rate mark, as the other two algorithms, however it is still the worse one.

The clean vs wrapped comparison also contextualises the Config A to Config B comparison from Section 4. Even without the clinical wrappers, Config B algorithms cannot reach Config A's performance ceiling. The continuous observation space itself, requiring generalisation across a 47-dimensional manifold rather than memorisation of 716 discrete values is a fundamental challenge independent of the clinical failure modes.

4. Configuration A vs Configuration B

4.1 State Space and Algorithm Feasibility

The fundamental difference is state representation. In Config A, 716 discrete states and 25 actions produce a Q-table of 17,900 entries, manageable in memory, with every entry updatable individually. The full MDP model enables model-based methods requiring no environment interaction.

In Config B, the 47-dimensional continuous observation makes tabular methods infeasible: the probability of revisiting an identical observation vector is effectively zero. Neural network function approximation becomes necessary, introducing new challenges: non-stationarity of learning targets, hyperparameter sensitivity, and the need for generalisation across unseen states. This is not merely a computational limitation but a qualitative shift, from memorising exact values to generalising across a continuous manifold.

4.2 Sample Efficiency

Policy Iteration in Config A converges to the exact optimum in 3 iterations with no environment interaction. The best model-free Config A method (QL Final) requires 400,000 episodes and still falls 1.6 pp short. In Config B, all algorithms require 1–2 million timesteps and cannot achieve Config A's relative improvement over baseline. This sample efficiency gap is fundamental: Config A's complete model eliminates exploration entirely; Config B's continuous observation and clinical wrappers require extensive interaction to learn a generalisable policy.

4.3 Performance Summary

Configuration	Random Baseline	Best Algorithm	Best Survival	Improvement
A — model-based	69.3%	Policy Iteration	84.1%	+14.8 pp
A — model-free	69.3%	QL Final (tuned)	82.5%	+13.2 pp
B — deep RL	64.6%	A2C 2M	71.0%	+6.4 pp

Table 6: Performance comparison across configurations.

The absolute improvement over baseline is larger in Config A (+14.8 pp) than Config B (+6.4 pp). However, Config B's lower baseline already reflects the additional challenge of clinical wrappers as acute events alone (~8% of episodes) represent irreducible stochasticity. The effective improvement range in Config B is therefore narrower than the raw numbers suggest.

4.4 Observability and Robustness

Config A presents a fully observable, clean environment, where each state index completely characterises the patient's clinical condition without noise or missing values. Config B introduces three forms of partial observability with no analogue in Config A. The clinical wrappers expose a critical vulnerability in naive deep RL. The transition from Config A to Config B thus represents not just a change in state representation but a qualitative shift in

deployment challenge, from a well-specified MDP solvable by classical methods to a partially observable, stochastic environment requiring robust policies.

Overall, the comparison demonstrates why tabular methods are highly effective in small, discrete, fully specified environments but become impractical as state spaces grow in dimensionality. Deep RL becomes necessary when observations are continuous and high-dimensional, yet this flexibility comes at the cost of reduced sample efficiency, greater computational requirements, and increased sensitivity to uncertainty.

5. Creative Extension: SOFA-Based Reward Shaping

5.1 Motivation

Standard RL for sepsis optimises only binary survival, treating every surviving trajectory as equally good regardless of the clinical path taken. A patient who survives after prolonged high organ dysfunction has experienced substantially worse outcomes than one who survives with consistently low SOFA scores. The SOFA (Sequential Organ Failure Assessment) score is a validated clinical metric for organ dysfunction severity (range 0–24) and a strong predictor of ICU mortality. It is available at every timestep in the observation vector, z-scored with a mean of −0.406 and a range of [−3.85, 1.05].

Binary survival also creates a gradient information problem. In episodes that end in patient death the agent receives a reward of zero for the entire episode, contributing no outcome gradient to the policy update regardless of how the patient's condition evolved during treatment. With the clinical wrappers active, approximately 30–35% of training episodes end in death. A per-step SOFA penalty converts every episode, including fatal ones, into a learning signal, reducing effective reward sparsity without changing the evaluation objective.

We investigate three variants. PPO is the unmodified baseline trained for 2M timesteps. PPO+Shaping adds the SOFA penalty ($\lambda = 0.005$) and raises ent_coef from 0.01 to 0.03, also trained for 2M timesteps. PPO+Combined stacks VecNormalize observation normalisation on top of PPO+Shaping, using the same hyperparameters and training budget. A sensitivity sweep over $\lambda \in \{0.001, 0.005, 0.01, 0.02\}$ with 1M timesteps each isolates the effect of penalty magnitude independently of the other modifications.

5.2 Implementation

We computed the reward by adding a small penalty proportional to the patient's current SOFA score encouraging the agent to keep the patient in a less critical state, not just survive (clinically motivated as SOFA correlates with organ failure). The shaped reward at each timestep is: $r_{shaped} = r - \lambda \times obs[25]$, wher obs[25] is the z-scored SOFA value. A positive value means the patient is sicker than average and generates a penalty; a negative value means the patient is healthier than average and generates a small reward bonus. The penalty is zero on average across the training distribution. At $\lambda = 0.005$ the mean per-step penalty magnitude is approximately 0.002, accumulating to roughly 0.018 over a typical nine-step episode — informative relative to a terminal survival reward of 1.0 but not dominant over it.

PPO+Shaping uses ent_coef = 0.03 instead of the baseline's 0.01. The denser shaped reward landscape causes faster early convergence and raises the risk of premature policy collapse; the higher entropy coefficient counteracts this by maintaining a stronger exploration incentive throughout training. All other hyperparameters are unchanged:

learning_rate = 3e-4, n_steps = 1024, 4 parallel environments, batch_size = 64, n_epochs = 10, clip_range = 0.2, gae_lambda = 0.95.

PPO+Combined applies VecNormalize to the shaped. VecNormalize computes running mean and variance statistics for each of the 47 features and standardises them online

5.3 Clinical Interpretation

The reward-shaping experiments provide insight into how clinically informed intermediate rewards influence learning behaviour in the sepsis environment. Both shaping variants achieved slightly higher survival rates than the standard PPO policy, increasing survival from 66.2% to 67.4% for PPO + Shaping and 67.1% for PPO + Combined. However, these gains were modest and were not accompanied by meaningful improvements in overall return, which remained similar across all three variants. This suggests that the baseline reward already provides a sufficiently informative optimisation signal and that additional SOFA-based shaping offers limited scope for further improvement.

The most notable effect of reward shaping appears in the robustness analysis. Under noisy observations, PPO + Shaping achieved the highest mean return (0.6344), outperforming both the standard PPO (0.5927) and PPO + Combined (0.6069). A similar pattern is observed under missing observations, where PPO + Shaping achieved a mean return of 0.6091 compared with 0.4933 for the standard PPO. PPO + Combined also improved robustness under missing observations (0.5993), although the benefit was slightly smaller than with shaping alone. These results suggest that incorporating organ dysfunction information into the reward function encourages policies that generalise more effectively when clinical information is incomplete or corrupted.

From a clinical perspective, this finding is potentially more important than the small improvement in overall survival. Real intensive care environments frequently contain missing measurements, delayed laboratory results, and noisy physiological observations. A policy that maintains performance under such conditions may be more practically deployable than one that achieves marginally higher performance only under ideal observability. The results therefore indicate that SOFA-based reward shaping primarily acts as a robustness-enhancing mechanism rather than a means of substantially increasing patient outcomes.

The relatively small differences between PPO, PPO + Shaping, and PPO + Combined also highlight an important limitation of reward engineering. While clinically informed shaping can influence learning behaviour and improve robustness, it does not fundamentally change the underlying challenges of the environment, including partial observability, stochastic acute events, and uncertainty in patient trajectories. Consequently, reward shaping should be viewed as a complementary tool for improving policy quality rather than a substitute for stronger algorithms or more informative state representations.

Algorithm	Survival	Mean Return	Noisy Eps Return	Missing Eps Return
PPO	66.2%	0.607	0.5927	0.4933
PPO + Shaping	67.4%	0.6061	0.6344	0.6091

Algorithm	Survival	Mean Return	Noisy Eps Return	Missing Eps Return
PPO + Combined	67.1%	0.6015	0.6069	0.5993

Table 7: Performance comparison across configurations (SOFA).

6. Conclusion

6.1 Summary of Findings

This project demonstrates the full arc from classical tabular RL to modern deep RL applied to a clinically grounded sepsis treatment optimisation problem.

In Configuration A, Policy Iteration achieves 84.1% survival using the exact MDP model, confirming it as the performance ceiling. Q-Learning progressively approaches this ceiling, from 76.1% (baseline) to 82.5% (tuned), with each improvement traceable to a principled algorithmic choice: optimistic initialisation for exploration, visit-based learning rates for convergence, and extended training for state coverage. The epsilon ablation confirms that exploration strategy has a first-order effect on final performance.

In Configuration B, A2C achieves the best performance (71.0% survival) through frequent on-policy updates suited to sparse reward signals. Its conservative, concentrated action distribution aligns with clinical best practice in sepsis management. PPO provides the most diverse treatment policies but is vulnerable to missing observations. DQN fails critically due to policy collapse under ε-greedy exploration, achieving only 5.5% action entropy and barely exceeding the random baseline, a fundamental limitation of ε-greedy exploration in sparse-reward environments.

The creative extension demonstrates that SOFA-based reward shaping can improve robustness to noisy and missing observations while having only a modest effect on overall survival. Compared with the unshaped PPO baseline, both shaping variants achieved substantially higher returns under missing observations, with PPO + Shaping improving mean return from 0.4933 to 0.6091 (approximately 24% relative improvement). However, gains in overall survival and mean return were small, suggesting that the primary benefit of reward shaping lies in improving policy robustness rather than maximising patient outcomes. These results indicate that clinically informed shaping rewards can help agents maintain performance when information is incomplete, a desirable property in real ICU settings where missing measurements are common.

6.2 Limitations

The ICU-Sepsis-v2 environment, while derived from real MIMIC-III data, remains a simulation with discretised patient trajectories and a simplified reward function. Real deployment would require out-of-distribution generalisation to new patient populations not represented in the training data. The SOFA observation in Config B is z-scored, meaning reward shaping operates on a normalised scale that may not correspond directly to clinical SOFA values. DQN's policy collapse could potentially be addressed with prioritised experience replay or a duelling.

Regarding hyperparameters, standard values were used instead of a grid or random search due to limited computational resources and time constraints, accepting the limitation of possibly suboptimal results. The same constraint influenced the number of training

timesteps: DQN was trained for 1M steps compared to 2M for the remaining algorithms, as it was significantly more expensive per step.

Annex

Figure 1. Policy Iteration Convergence

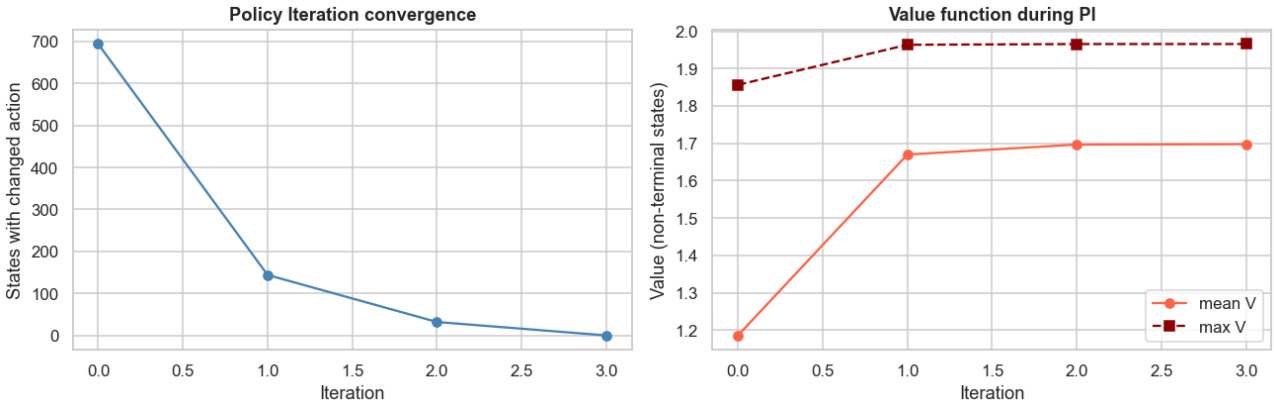


Figure 2. Survival Rate PI vs QL

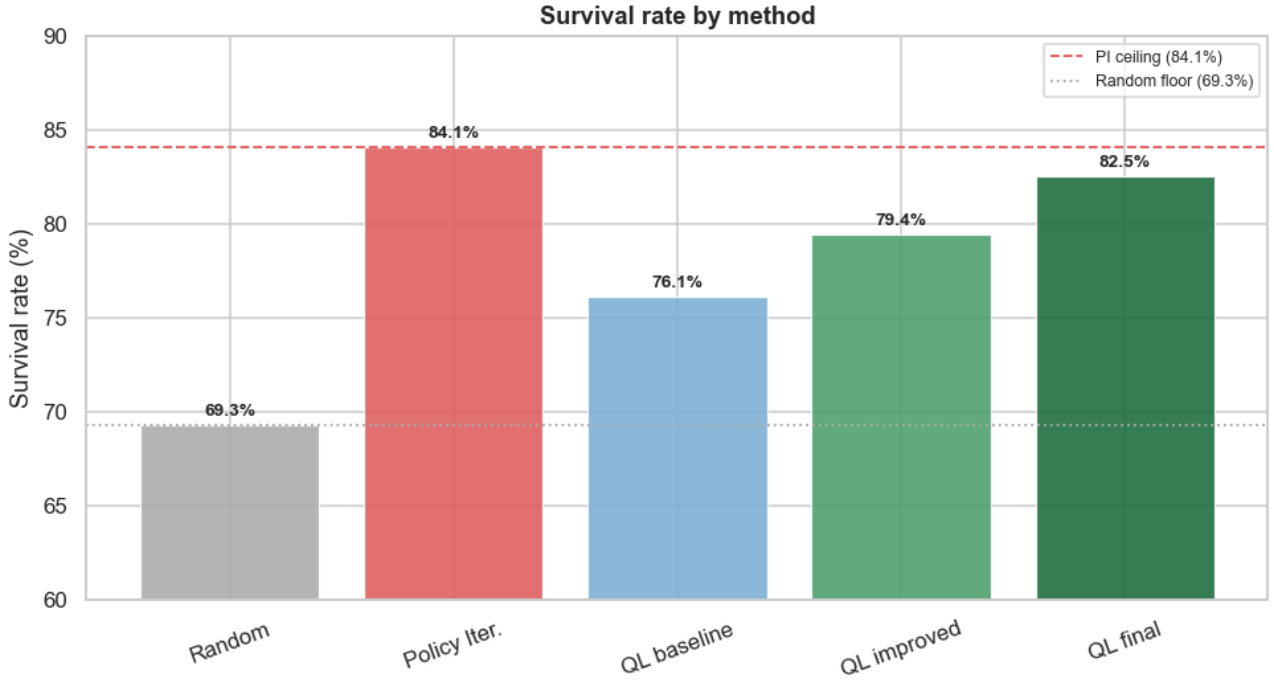


Figure 3. Policy Agreement

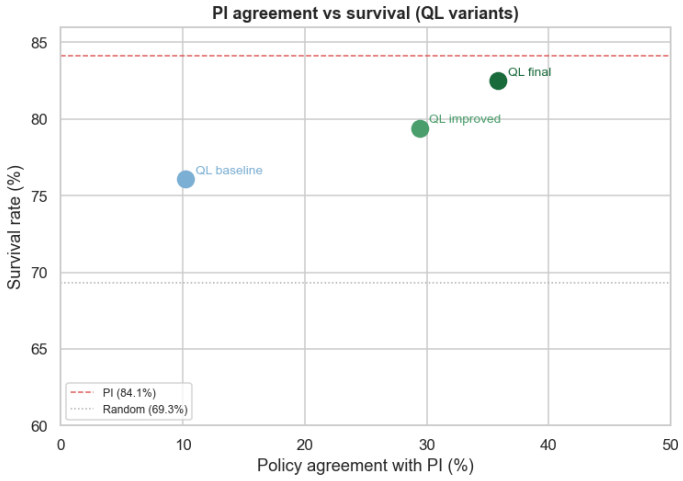


Figure 4. QL Learning Curves

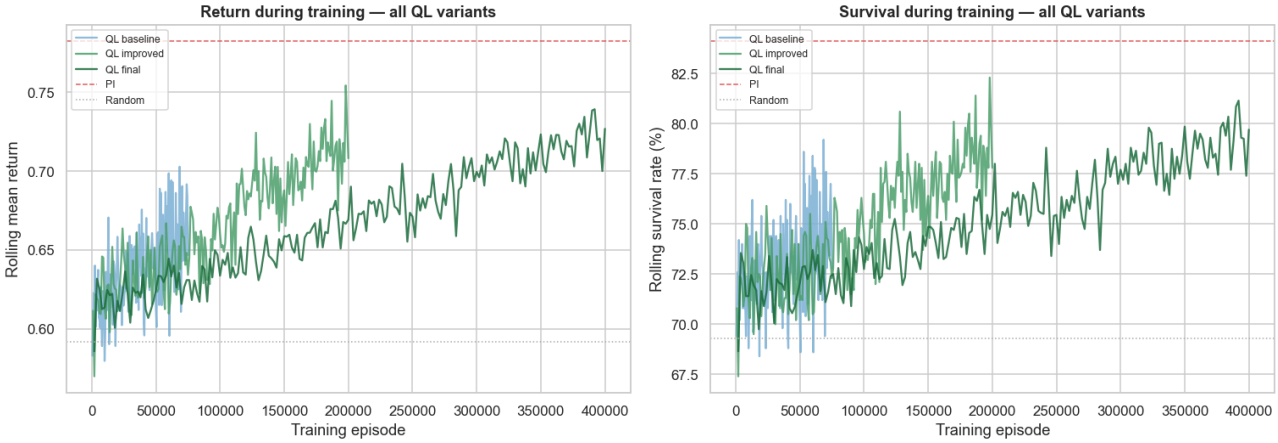


Figure 5. Treatment Intensity

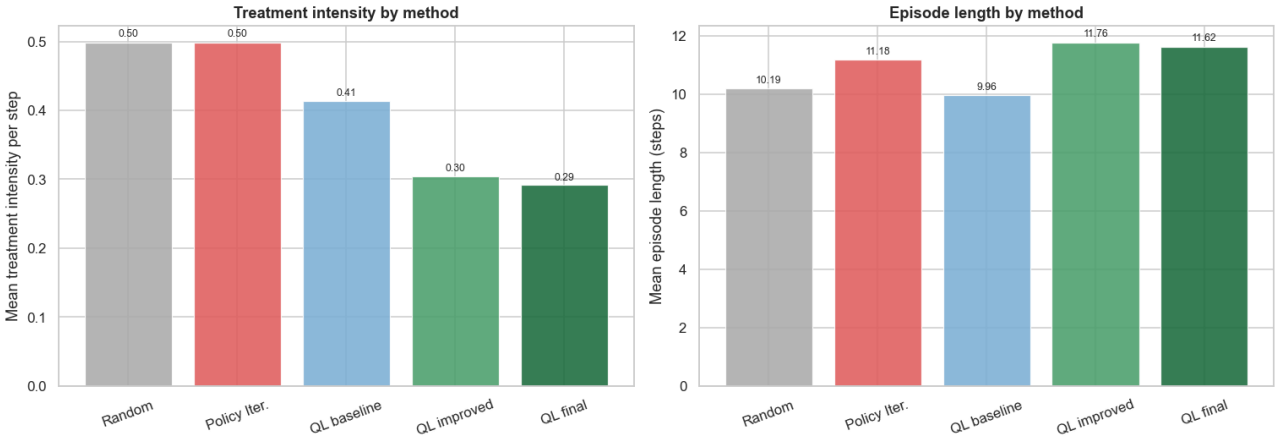


Figure 6. Exploration vs Exploitation QL



Figure 7. PI vs QL Action Policy

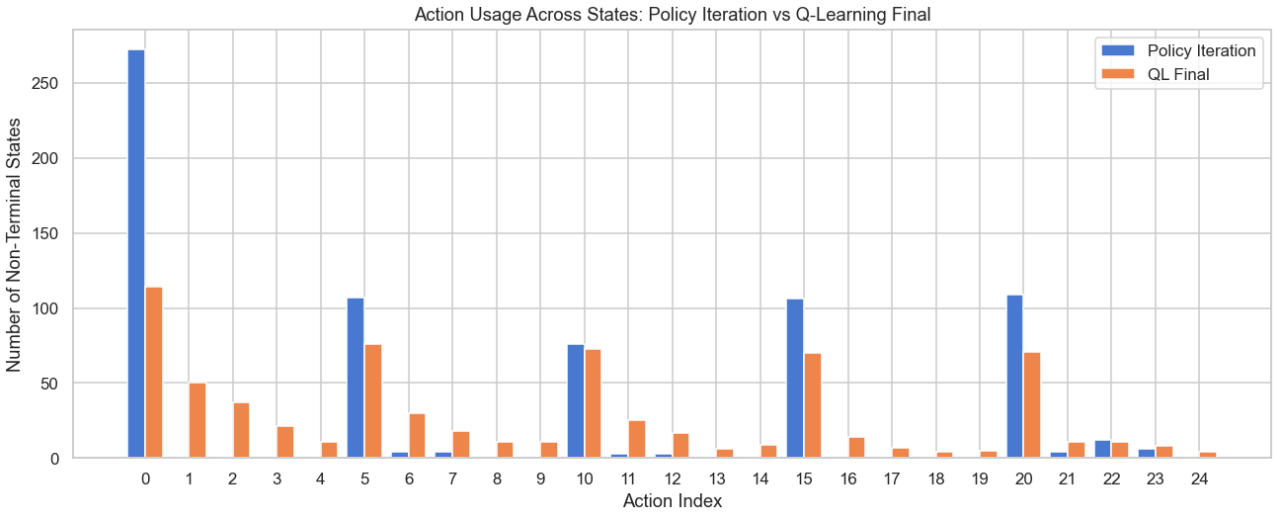


Figure 8. Config B Training Curves

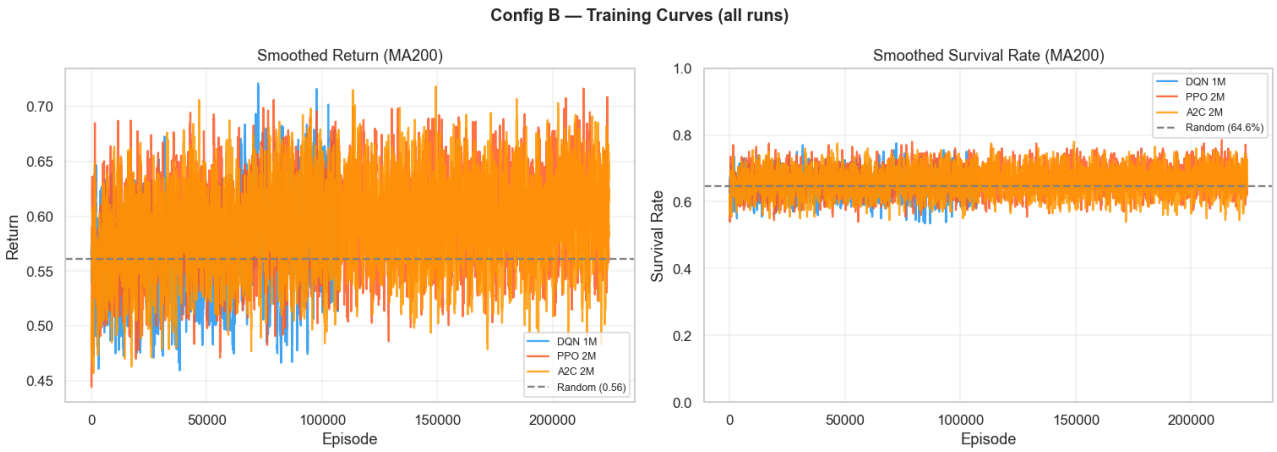


Figure 9. Config B Convergence

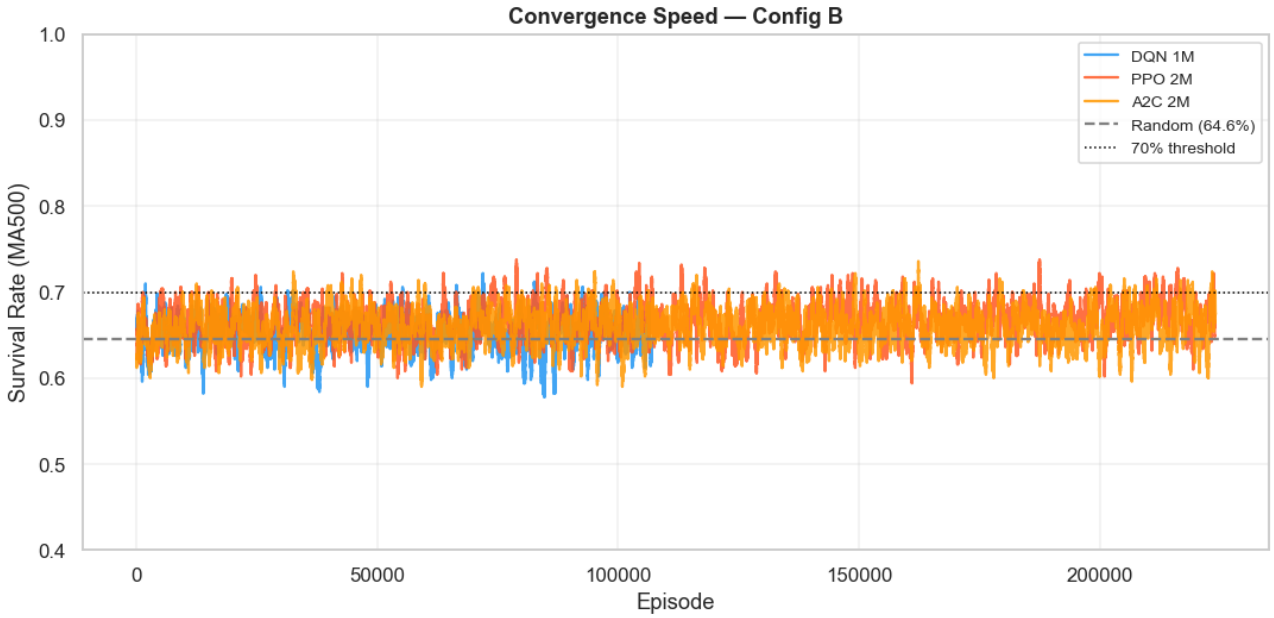


Figure 10. Config B Action Distribution

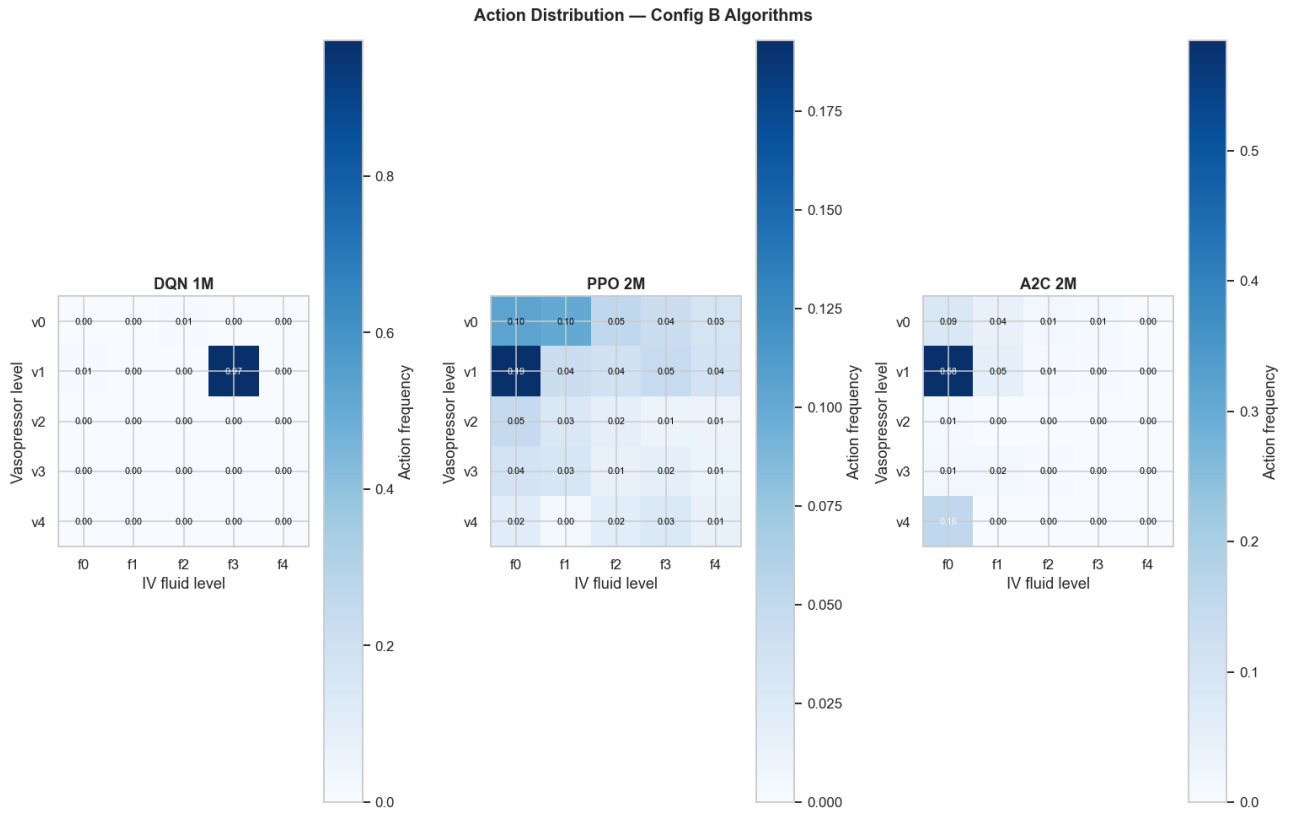


Figure 11. Config B Exploration vs Exploitation

